QHO1D Spatial Truncation

Finite-box and finite-difference technical report

Project Koios

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Status. This is the maintained calculation-facing technical report for `calculations/qho1d`. It is not sealed evidence, a publication manuscript, a continuum-convergence proof, or a physical uncertainty statement. The plotted arrays are authenticated binary64 development outputs. Exact Lean results are cited separately and do not certify the runtime arrays in this experiment.

Abstract

This report studies spatial truncation: how a one-dimensional quantum harmonic oscillator on the real line changes when represented on a finite Dirichlet box and grid, then compared with a retained number-state reference. The experiment compares a centered second-order finite-difference Hamiltonian with the direct diagonal number-state Hamiltonian after sampled Hermite states have been Gram-orthonormalized and used as a common-coordinate injection. It records Frobenius, diagonal, off-diagonal, Gram, isometry, refinement, eigenvalue, eigenstate, residual, and adjacent-grid diagnostics. The direct sweep demonstrates the expected second-order spatial trend once the finite box is sufficiently large, while a smaller box exhibits a distinct boundary-dominated regime. Formal results establish the exact Hermite–Gaussian pointwise eigenfunction identity, finite ladder boundary defect, compression leakage identity, and conditional common-coordinate algebra. They deliberately do not establish the remaining finite-box, discretization, or floating-point error bounds.

1 Question and scope

The computational question is not whether the harmonic oscillator is analytically solvable. It is how several finite representations differ when they are compared in one explicitly constructed coordinate space. For box half-width b, realized grid spacing h, and retained Python dimension K, the experiment constructs

$$D_{b,h,K} = J^{\mathsf{T}} H_{b,h} J - H_K. \tag{1}$$

Here $H_{b,h}$ is the finite Dirichlet-grid Hamiltonian, H_K is the direct diagonal reference with entries $n + \frac{1}{2}$ for $n = 0, \ldots, K-1$, and J maps retained number-state coordinates into the interior grid.

The experiment is nondimensional with $\hbar = m = \omega = \ell = 1$. It does not claim that a finite grid is the continuum operator, that the selected sweep proves convergence, or that the discrepancy is a single uncertainty estimate. Its purpose is to expose separately measurable error channels and to provide reusable diagnostics for later certified bounds.

2 Representations and comparison map

Let x_i denote the ordered interior coordinates of a uniform grid on $[-b, b]$ with homogeneous Dirichlet endpoints. The spatial matrix uses the centered stencil

$$(\Delta_h f)_i = \frac{f_{i-1} - 2f_i + f_{i+1}}{h^2}, \qquad H_{b,h} = -\frac{1}{2}\Delta_h + \frac{1}{2}\operatorname{diag}(x_i^2). \tag{2}$$

The real-line spectral problem and its Hermite-state solution are standard [3, 5]. The exact real-line number-state samples are

$$\psi_n(x) = \frac{H_n(x)e^{-x^2/2}}{\pi^{1/4}\sqrt{2^n n!}}, \qquad E_n = n + \frac{1}{2}. \tag{3}$$

The implementation evaluates the normalized states through their stable three-term recurrence. With $S_{in} = \sqrt{h}\,\psi_n(x_i)$, it constructs

$$G = S^{\mathsf{T}}S, \qquad R = G^{-1/2}, \qquad J = SR. \tag{4}$$

The inverse square root is computed from a binary64 symmetric eigendecomposition using standard finite matrix methods [2]. It is therefore checked numerically; it is not the exact inverse square root assumed by the corresponding Lean isometry theorem.

The direct diagonal definition of H_K is intentional. Constructing the Hamiltonian from products of already-truncated ladder matrices introduces a highest-state boundary defect even before spatial discretization is considered.

3 Formal crosswalk

The proof sidecar uses Lean 4 and mathlib [1, 6] in the `projectkoios-formal` repository at local commit

`002e1a2b4167acf9fe467a4f3da733c23dfd28e1`.

The following results are used as mathematical boundaries rather than runtime certificates.

The formal cutoff K_F labels the highest retained state and has dimension $K_F + 1$. The Python field `retained_dimension` is the dimension K and contains states 0 through $K-1$. Thus $K_F = K-1$ for every nonempty Python case. The audited Lean proofs contain no `sorry`; their reported transitive foundations are limited to the repository's disclosed standard classical axioms. No theorem binds the exact matrices directly to the JSON bytes produced here.

Formal identifier	Relevance to this experiment
`QHO-HERMITE-THM-001`	The constructed physicists' Hermite polynomials satisfy the exact differential equation.
`QHO-HERMITE-THM-002`	The unnormalized Hermite–Gaussian satisfies the pointwise dimensionless oscillator eigenfunction identity with eigenvalue $n + \frac{1}{2}$. This is not an L^2 completeness theorem.
`QHO-LADDER-THM-001`	A retained ladder commutator equals $I - (K_F + 1)\|K_F\rangle\langle K_F\|$ in formal dimension $K_F + 1$.
`QHO-TRUNCATION-THM-001`	Compression fails to preserve products by the exact omitted-space leakage term $J^{\mathsf{T}} A(I - JJ^{\mathsf{T}})BJ$.
`QHO-TRUNCATION-THM-002`	Reconstructing the oscillator Hamiltonian from truncated ladder products subtracts an explicit highest-state projector term.
`QHO-COMPARISON-THM-001`	If $R^{\mathsf{T}}(S^{\mathsf{T}}S)R = I$ exactly, then $J = SR$ is exactly isometric.
`QHO-COMPARISON-THM-002`	The squared finite-matrix Frobenius magnitude splits exactly into diagonal and off-diagonal contributions.

Table 1: Exact proof results and their deliberately bounded applicability.

4 Direct calculation protocol

The declared Cartesian sweep is

$$b \in \{4, 6, 8\}, \qquad h_{\text{requested}} \in \{0.2, 0.1, 0.05\}, \qquad K \in \{2, 4, 6\}. \tag{5}$$

This produces 27 deterministically ordered cases. Each case retains the realized spacing, interior-point count, Gram matrix, inverse Gram square root, injection, pulled-back Hamiltonian, direct reference, signed difference, scalar discrepancy channels, and the full-space residual norm.

The implementation is library-owned by `ksdft2effmass.model_systems.qho1d`. The calculation directory owns thin direct entry points, the authored protocol, execution paths, and this report. The finite difference, ladder, typed-quantity, shared JSON, verification, and PNG mechanisms remain with their reusable owners.

5 Eigenstates and finite-grid distortion

Figure 1 compares the first four real-line Hermite states against the sign-aligned eigenvectors of the finite Dirichlet-grid Hamiltonian for $b = 8$, $h = 0.05$, and $K = 6$. A Euclidean-normalized grid eigenvector v_n is displayed as $v_n/\sqrt{h}$ so that $h \sum_i |v_{n,i}/\sqrt{h}|^2 = 1$. The sign is chosen only for visual comparison and carries no physical content.

Figure 2 exposes the difference hidden by the overlay. For states $n = 0, 1, 2, 3$, the grid-weighted state-difference norms are approximately 1.15×10^{-4}, 3.27×10^{-4}, 6.53×10^{-4}, and 1.11×10^{-3}. The finite-grid energy errors for $n = 0, \ldots, 5$ are

$$(-7.81\times10^{-5}, -3.91\times10^{-4}, -1.02\times10^{-3}, -1.95\times10^{-3}, -3.21\times10^{-3}, -4.77\times10^{-3}). \tag{6}$$

These are development observations for one selected grid, not spectral enclosures.

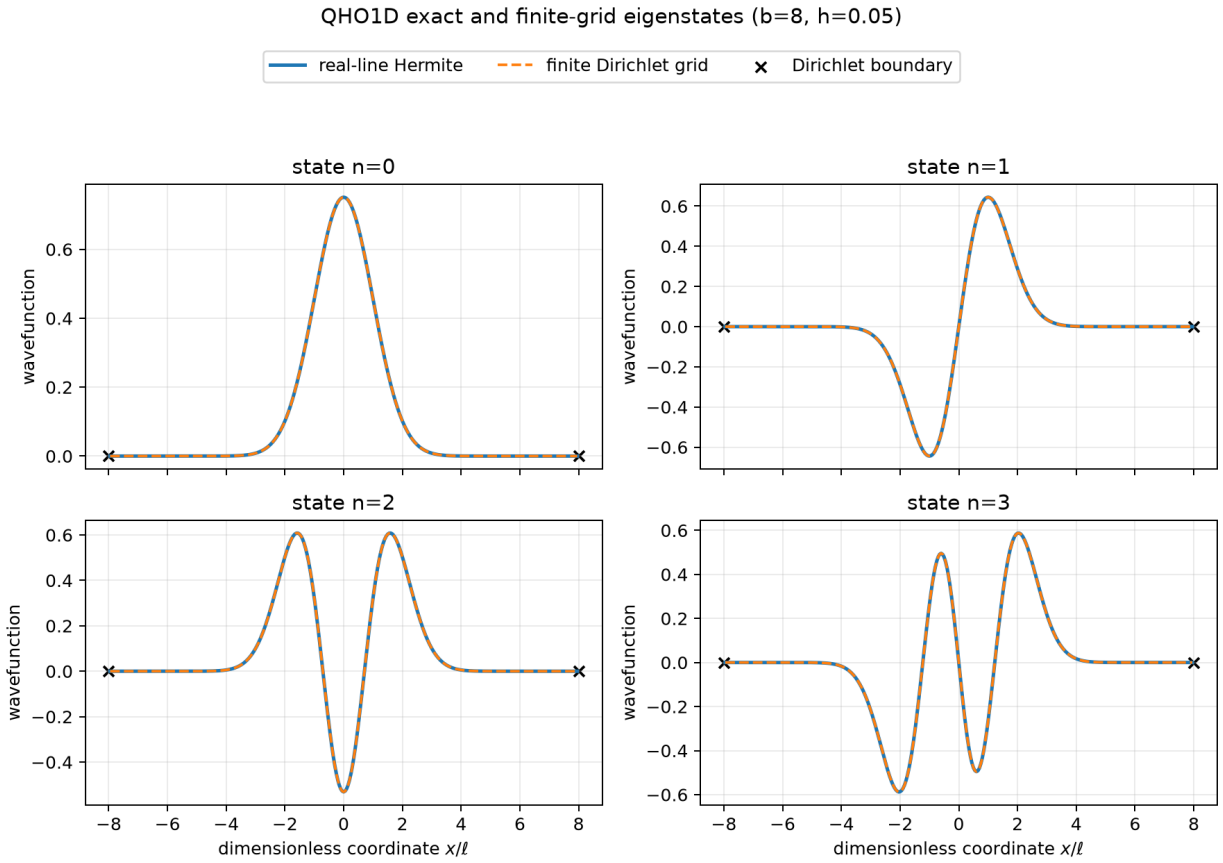


Figure 1: Real-line Hermite states and finite-grid eigenstates for the selected fine case. The agreement at plot scale does not replace the distortion diagnostics.

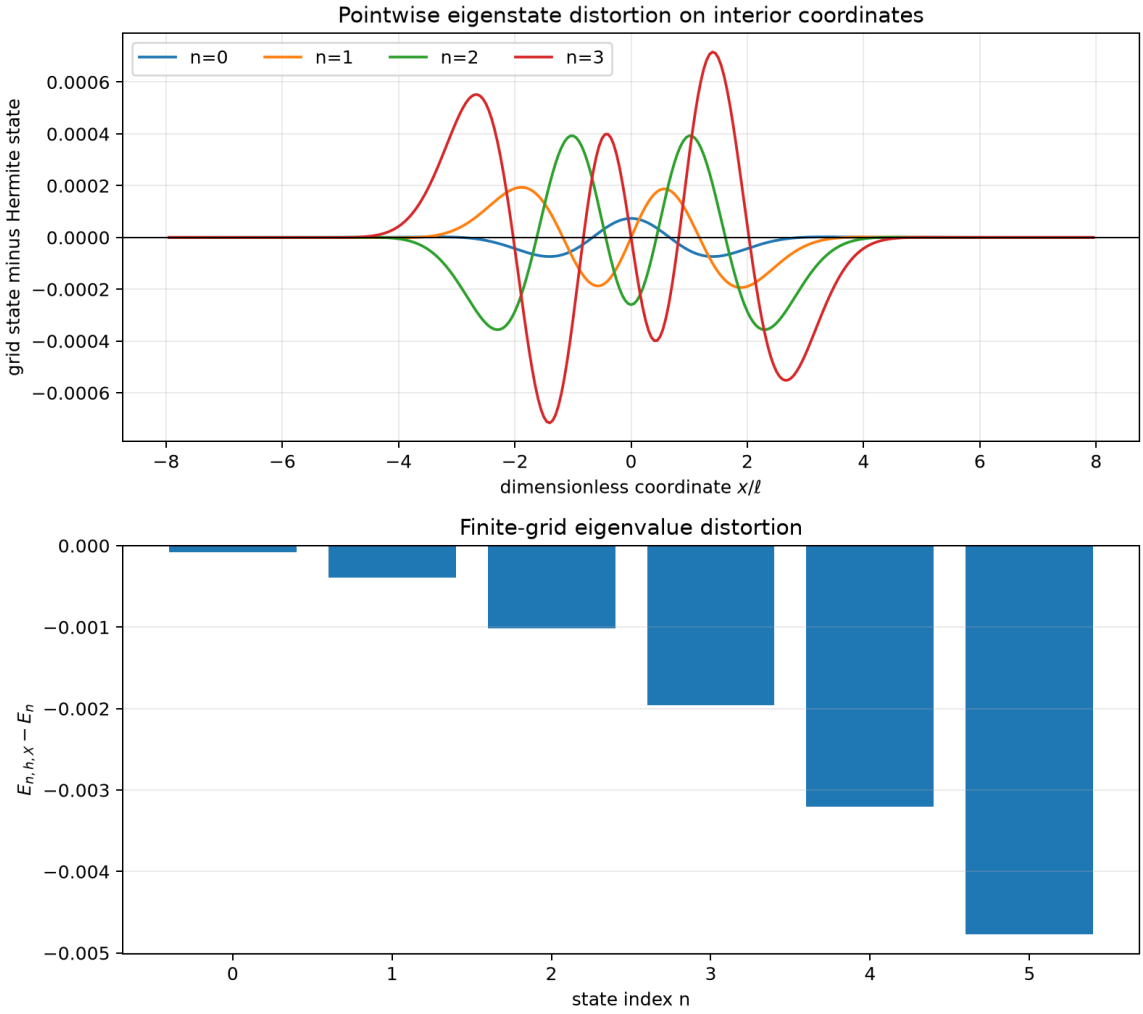


Figure 2: Pointwise state distortion and finite-grid eigenvalue distortion for $b = 8$, $h = 0.05$, and the first six eigenvalues.

6 Reusable diagnostics

For each case the result records

$$\delta_{\mathrm{abs}} = \|D_{b,h,K}\|_{\mathrm{F}}, \qquad \delta_{\mathrm{rel}} = \frac{\|D_{b,h,K}\|_{\mathrm{F}}}{\|H_K\|_{\mathrm{F}}}, \tag{7}$$

$$\delta_G = \|G - I\|_{\mathrm{F}}, \qquad \delta_J = \|J^{\mathsf{T}} J - I\|_{\mathrm{F}}. \tag{8}$$

It also records $\|\operatorname{diag} D\|_{\mathrm{F}}$ and $\|D - \operatorname{diag} D\|_{\mathrm{F}}$. These channels distinguish normalization and common-coordinate effects from diagonal energy shifts and representation-induced couplings.

A further reusable a-posteriori diagnostic is

$$\rho_{b,h,K} = \|H_{b,h} J - J H_K\|_{\mathrm{F}}. \tag{9}$$

For an exact isometry $J^{\mathsf{T}} J = I$,

$$J^{\mathsf{T}} H_{b,h} J - H_K = J^{\mathsf{T}} (H_{b,h} J - J H_K), \tag{10}$$

so $\delta_{\mathrm{abs}} \le \rho_{b,h,K}$ in the Euclidean operator/Frobenius bound. The plot reports the binary64 evaluation of both quantities. It is not an interval-certified inequality.

6.1 Adjacent-grid change in common coordinates

The direct typed result also supports comparison between adjacent authored grid spacings at fixed b and K. The parent finite-grid matrices have different spatial dimensions and are not subtracted. Instead, define

$$\chi_{b,K}(h_{\mathrm{c}}, h_{\mathrm{f}}) = \left\| J^{\mathsf{T}}_{b,h_{\mathrm{c}},K} H_{b,h_{\mathrm{c}}} J_{b,h_{\mathrm{c}},K} - J^{\mathsf{T}}_{b,h_{\mathrm{f}},K} H_{b,h_{\mathrm{f}}} J_{b,h_{\mathrm{f}},K} \right\|_{\mathrm{F}}, \qquad h_{\mathrm{c}} > h_{\mathrm{f}}. \tag{11}$$

Both pullbacks act in the same ordered retained number-state coordinates, so the subtraction is defined. `Qho1dCrossGridAnalyzer` evaluates the norm through the canonical PhysKit `FrobeniusMatrixNorm`. Its paired verifier checks the authored controls, realized spacings, represented dimensions, and norm directly from the typed case matrices. It does not reconstruct those matrices; that independent route remains owned by `Qho1dVerifier`.

Table 2 reports the six adjacent-grid observations at $b = 8$. The coarse-to-fine decrease is close to a factor of four over this bounded sequence, consistent with the refinement trend already visible in the relative discrepancy. The values are changes between two finite representations, not errors against an unknown continuum operator.

K	h_{c}	h_{f}	$\chi_{8,K}(h_{\mathrm{c}}, h_{\mathrm{f}})$
2	0.20	0.10	4.7528×10^{-3}
2	0.10	0.05	1.1934×10^{-3}
4	0.20	0.10	2.8909×10^{-2}
4	0.10	0.05	7.2795×10^{-3}
6	0.20	0.10	8.4552×10^{-2}
6	0.10	0.05	2.1362×10^{-2}

Table 2: Adjacent-grid pulled-back Hamiltonian changes at fixed $b = 8$.

The complete configured inventory contains 18 adjacent-grid comparisons. The post-analysis verifier reports 90 exact derivation checks: five controls or values for each pair. Passing those checks establishes derivation from the supplied direct-result matrices only; it does not create a new independent continuum reference.

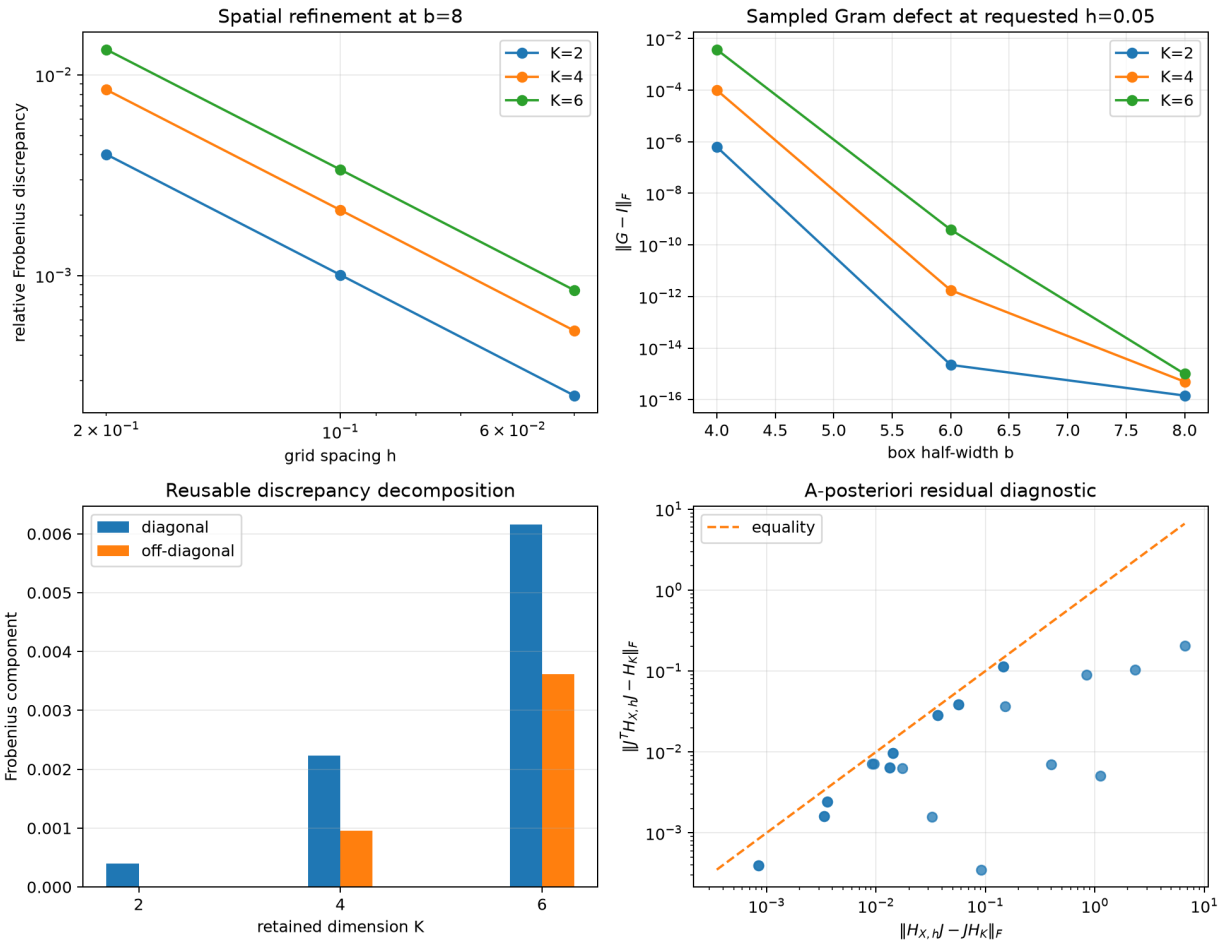


Figure 3: Spatial refinement, sampled Gram defect, diagonal/off-diagonal decomposition, and residual diagnostic. Every panel reports authenticated binary64 development data.

7 Observed regimes

Table 3 shows the relative discrepancy for the largest retained space. At $b = 6$ and $b = 8$, halving h reduces the discrepancy by factors close to four, consistent with the expected centered-difference order over this limited sweep. This is an observed trend, not a proof or fitted asymptotic bound. Results for $b = 6$ and $b = 8$ are nearly identical at the displayed scale, indicating that the real-line tails are below the leading discretization effect for these cases.

At $b = 4$, refinement does not improve the combined discrepancy and ultimately makes it larger. The sampled Gram defect is also much larger. This is useful rather than pathological: it demonstrates that grid refinement cannot be interpreted without a separate finite-box channel. A finer stencil resolves the fixed, boundary-incompatible box problem more accurately; it does not turn that problem into the real-line oscillator.

For the selected $(b, h, K) = (8, 0.05, 6)$ case, the absolute discrepancy is 7.1407×10^{-3}; its diagonal and off-diagonal components are 6.1581×10^{-3} and 3.6150×10^{-3}, respectively. Their squared values, not their values, add to the squared total as established by the exact finite-matrix decomposition. The Gram defect is 9.93×10^{-16} and the measured injection isometry error is 3.22×10^{-15}, both at binary64 scale for this case.

b	h	K	δ_{rel}
4	0.20	6	1.0570×10^{-2}
4	0.10	6	1.2283×10^{-2}
4	0.05	6	2.4372×10^{-2}
6	0.20	6	1.3370×10^{-2}
6	0.10	6	3.3708×10^{-3}
6	0.05	6	8.4448×10^{-4}
8	0.20	6	1.3370×10^{-2}
8	0.10	6	3.3708×10^{-3}
8	0.05	6	8.4448×10^{-4}

Table 3: Relative common-coordinate Frobenius discrepancy for $K = 6$.

8 Error-channel ledger

The observed discrepancy must not be collapsed into one unnamed error. The current ledger is:

1. **Exact finite ladder truncation.** Truncated ladder products have an explicit top-state boundary defect. The Python comparison avoids this channel by using the direct diagonal reference.

2. **Finite-box closure.** Real-line Hermite states do not exactly satisfy a finite homogeneous Dirichlet boundary. The $b = 4$ observations show that this channel can dominate refinement.

3. **Interior discretization.** The centered stencil has the standard local second-order consistency mechanism [4], but this report does not supply a governed global constant or regularity-to-spectrum theorem.

4. **Sampled-state and Gram correction.** Finite quadrature changes the Gram matrix, and numerical inverse-square-root normalization changes the sampled columns.

5. **Projection and pullback.** $J^{\mathsf{T}} H_{b,h} J$ measures the operator in one retained coordinate space; it is not the full grid spectrum.

6. **Eigensolver and floating point.** Every runtime matrix, eigendecomposition, norm, and serialized decimal is binary64 and has no interval certificate here.

9 Reproduction and provenance

From the repository's `python/` directory, the direct development lifecycle is

```
PYTHONPATH=$PWD/src uv run python ../calculations/qho1d/run_01_calculation.py
PYTHONPATH=$PWD/src uv run python ../calculations/qho1d/run_02_verification.py
PYTHONPATH=$PWD/src uv run python ../calculations/qho1d/run_03_render.py
```

The execution is written beneath `calculations/qho1d/executions/direct-spatial-truncation-3x3x3-section`. The typed result groups every case by spatial representation, sampled-state alignment, Hamiltonian comparison, and residual. Its verification record, visualization projection, and rendering manifest bind exact SHA-256 identities. Those identities establish byte correlation only; the execution remains unsealed development output unless a separately authorized seal is created.

The adjacent-grid table is a typed post-analysis of that retained result. It does not rewrite the execution result or add a stage to the execution manifest. It is reproducible by decoding `results/result.json`, executing `Qho1dCrossGridAnalyzer`, and passing the returned aggregate with the decoded result to `Qho1dCrossGridVerifier`.

10 Next steps

The next useful result is a channel-separated bound, not a larger undifferentiated sweep. Priorities are to retain boundary-row and interior residuals separately, serialize the full residual diagnostic, establish a governed residual-to-spectrum statement, derive an explicit finite-box Hermite-tail estimate, and connect the centered-difference consistency residual to a global eigenvalue/eigenstate bound. Interval or otherwise checked floating-point certificates would be required before promoting binary64 observations to rigorous numerical enclosures.

References

[1] Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction—CADE 28*, volume 12699 of *Lecture Notes in Computer Science*, pages 625–635, Cham, 2021. Springer.

[2] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins Studies in the Mathematical Sciences. Johns Hopkins University Press, Baltimore, MD, 4 edition, 2013.

[3] Ramamurti Shankar. *Principles of Quantum Mechanics*. Plenum Press, 2 edition, 1994.

[4] John C. Strikwerda. *Finite Difference Schemes and Partial Differential Equations*. Society for Industrial and Applied Mathematics, Philadelphia, PA, 2 edition, 2004.

[5] Gerald Teschl. *Mathematical Methods in Quantum Mechanics: With Applications to Schrödinger Operators*, volume 157 of *Graduate Studies in Mathematics*. American Mathematical Society, Providence, RI, 2 edition, 2014.

[6] The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs*, pages 367–381, New York, NY, USA, 2020. Association for Computing Machinery.